

Lakshya Pant

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Personal Profile

I am a motivated Mechanical Engineering undergraduate with hands-on experience in CAD design, CFD simulation, and mechanical system analysis. Proficient in AutoCAD, SolidWorks, ANSYS, MATLAB, SpaceClaim, FEA, CFD, and Simulink, and I enjoy solving real-world problems through a mix of research, manual engineering, and software simulations. With leadership experience in the IMechE Railway Challenge and STEM Society, I thrive in collaborative environments and am eager to contribute my creativity and technical skills to a dynamic engineering role.

Education

BEng (Hons) Mechanical Engineering (Predicted 2:1) 2024 - 2026

Anglia Ruskin University, Chelmsford

Relevant Modules: Thermodynamics, Automation and Control, CAD Modelling, Engineering Mathematics, Materials and Manufacturing, Mechanical Systems, Fluid Mechanics

Awards & Achievements

- Completed ARU Research & Professional Ethics Certification
- Achieved high academic performance in CAD and simulation-based modules
- Selected to join the team representing ARU at the IMechE and European Railway Challenge
- Best Design Poster Award, ARU Engineering End-of-Year Show 2026, awarded for outstanding poster design in Design of Drive System for Railways.

Engineering Experience

IMechE Railway Challenge – Team Heron Project Member Aug 2025 – Present

- Produced engineering drawings and 3D components using AutoCAD
- Supported design and evaluation of railway vehicle drive systems
- Optimised design for power transmission, durability, and energy efficiency

Structural Analysis for Wind Turbine Blade Jan 2026 – May 2026

- Conducted structural analysis for a wind turbine blade using Qblade Software for aerodynamic performance evaluation
- Applied fatigue analysis techniques

Automotive Air-Duct System Design & Simulation Project Aug 2025 – Dec 2025

- Designed an HVAC air-duct system in CAD with efficient flow routing
- Optimised duct geometry to improve airflow smoothness, split balance and outlet uniformity
- Verified performance using ANSYS, CFD and Simulink to meet pressure-drop and thermal targets

Drone Design and Optimisation Project Jan 2025 – May 2025

- Applied DFMA principles using Boothroyd-Dewhurst DFMA software to improve manufacturability and cost efficiency
- Performed FEA simulations using ANSYS to assess the structural integrity and fatigue performance of a drone.
- Designed innovative lock in mechanism for the drone

Heat Transfer Analysis Jan 2025 – May 2025

- Simulated 2D steady-state heat conduction through a carbon steel rectangle using MATLAB and the Finite Difference Method to solve the heat conduction equation.
- Analysed the impact of mesh size and boundary conditions on simulation accuracy

Joystick Component Design & Manufacturing Project Jan 2024 – May 2024

- Designed a joystick mechanism with controlled 45-degree motion in CAD
- Manufactured components using milling and lathe equipment

Strawberry Cutter Design Project

Sept 2024 – Dec 2024

- Designed and modelled a strawberry cutter mechanism, focusing on X and Y axis movements
- Developed CAD models to optimise blade travel and cut accuracy

Leadership Roles

STEM Society Treasurer & Hackathon Organiser

Aug 2024 – Present

Anglia Ruskin University, Chelmsford

- Managing budgets and allocating resources for engineering projects and events
- Coordinated finances, logistics, event schedules, roles and responsibilities to ensure smooth delivery of annual engineering hackathons
- Maintained financial transparency and accurate documentation
- Collaborated with committee members to deliver a successful, well-attended student event

Chelmsford Science and Engineering Festival Volunteer

Oct 2024

- Supported delivery of STEM workshops and public engagement activities
- Assisted with event setup, coordination, and visitor engagement
- Communicated scientific and engineering concepts to a non-technical audience

Work History

Bar Team Member at Stonegate Group, Chelmsford

2024 - Present

- I work effectively as part of a team to ensure excellent customer service in a fast-paced environment, demonstrating and developing strong organisation, communication, interpersonal, multitasking and time management skills.

Key Skills

Technical: AutoCAD, MATLAB, Simulink, ANSYS (FEA, CFD), Autodesk Inventor, Fusion 360, SolidWorks, Qblade, Ansys SpaceClaim, Ansys Discovery, 3D CAD modelling, engineering drawings, manufacturing documentation

IT Skills: Excel (Advanced), PowerPoint, Microsoft Word, Microsoft Project

Engineering: Mechanical design, manufacturing design, HVAC and airflow analysis, drivetrain systems, structural analysis, DFMA, CAD-to-manufacture workflows

Simulation & Analysis: End-to-end CFD/FEA simulation setup, advanced boundary condition definition, mesh refinement, result interpretation, process data analysis, result documentation

Analytical: Problem solving, data analysis, test result interpretation, design evaluation

Professional: Technical reporting, engineering documentation, project budgeting, stakeholder engagement, regulatory compliance, positive attitude, time management

Leadership & Teamwork: Team collaboration, project coordination, event organisation

Languages: English (Fluent)

Personal Interests

- Mechanical design and engineering simulation
- Exploring new engineering tools and technical innovations
- Fitness, gym training and football

References Available on Request
